Supplementary material for: Second-order MCSCF optimization revisited. I. Improved algorithms for fast and robust second-order CASSCF convergence

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I. PSEUDO CODES FOR THE OPTIMIZATION ALGORITHMS

The following pseudo codes show the various default parameters and thresholds used in the optimization algorithms.

A. Uncoupled (CI)

```
One orbital optimization step  → |ΔR|

if |ΔR|>0.1 then
  N_micro = 5
  N_orb = 4
else if |ΔR|>0.01 then
  N_micro = 7
  N_orb = 3
else
  N_micro = 20
  N_orb = 1
end if

N_orb-1 orbital optimization steps

do microiterations = 1,N_micro-1
  CI Davidson optimization (two steps)

  N_orb orbital optimization steps

  if convergence → exit loop
end do
```

B. Uncoupled (Orb)

```
do microiterations = 1,20
  CI Davidson optimization
    (reduce the CI residual by 0.3)

  One orbital optimization step

  if convergence → exit loop
end do
```

C. Coupled

```
do microiterations = 1,20
  CI Davidson optimization → reduced space C
    (the CI residual reduced by factor 0.3)
  build h̃_cc, h̃_co ,h̃_oo, g̃_c, g̃_o
  do k = 1,N_add (N_add=5)
    AH method → S^k, ΔR^k, E^k (eq. 44)
    if k>1 and |(E^k-E^{k-1})/E^k)|<ε_add (ε_add=0.05) then
      exit loop
    end if
    if "Partially Coupled"  → exit loop
    build coupled residuum with S^k, ΔR^k (eq. 53)
    add expansion vector to C
    update h̃_cc,h̃_co,g̃_c
  end do
  CI update with exp(S^k)
  update U,T with exp(ΔR^k)
  if convergence → exit loop
end do
```

D. QN-Coupled

Start with the Uncoupled (Orb) optimization

if $|\Delta\mathbf{R}| < \epsilon$ (ϵ=0.05) $\rightarrow$ activate QN optimization, store N_{micro}

QN optimization:

$\mathbf{T}^0 = \mathbf{T}$ from Two-Step

$k = 0$

do microiterations = 1,20-N_{micro}

  CI Davidson optimization

    if k>=1 Davidson until $|\mathbf{r}_{ci}| < 0.1|\tilde{\mathbf{g}}^{k-1}|$

  calculate gradient with the new density $\mathbf{D}^k \rightarrow \tilde{\mathbf{g}}^k$

  evaluate energy $E^k = E^{(2)}(\mathbf{D}^k, \mathbf{T}^k)$

  if k==0 or $|\mathbf{D}^k - \mathbf{D}^{k-1}| > \epsilon_c$ (ϵ_c=0.025) or $|E^k - E^{k-1}| > 0$ then

    orbital Hessian with $\mathbf{T}^k \rightarrow \tilde{\mathbf{h}}^k$

  end if

  if k>=1 and $|E^k - E^{k-1}| < 0$ then

    QN update of the Hessian (eq. 59) with:

      $\mathbf{y}^k = \tilde{\mathbf{g}}^k - \tilde{\mathbf{g}}^{k-1}$ and $\mathbf{s}^k = \Delta\mathbf{R}^{k-1} \rightarrow \tilde{\mathbf{h}}^k$

  end if

  AH method with $\tilde{\mathbf{g}}^k$, $\tilde{\mathbf{h}}^k \rightarrow \Delta\mathbf{R}^k$

  Update $\mathbf{T}^k \rightarrow \mathbf{T}^{k+1}$

  $k = k + 1$

  if convergence $\rightarrow$ exit loop

end do

E. Microiteration convergence criterion

ϵ_{acc} = min(0.01|$\tilde{\mathbf{g}}^0$|,|$\tilde{\mathbf{g}}^0$|^2)

i = microiteration

if

  total gradient sqrt(|$\tilde{\mathbf{g}}^i$|^2+|$\tilde{\mathbf{r}}^i_{ci}$|^2) < ϵ_{acc}

  or

  |$\tilde{\mathbf{g}}^i$| < 0.1ϵ_{acc}

  or

  |E^i-E^{i-1}| < 0.01ϵ_{acc} and i>=3

then

  convergence reached

end if

II. GEOMETRIES AND ENERGIES OF MOLECULES IN SECTION D

Coordinates in Angstrom, energies in Hartree. 1 Angstrom = 1.88972612 Bohr

A. Co2(CO)6C2H2

Co	0.0000000000	1.2342550000	0.1406580000
Co	0.0000000000	-1.2342550000	0.1406580000
C	0.6765240000	0.0000000000	1.5563930000
C	-0.6765240000	0.0000000000	1.5563930000
H	1.5499390000	0.0000000000	2.1895500000
H	-1.5499390000	0.0000000000	2.1895500000
C	-1.4607320000	1.5258320000	-0.8965450000
O	-2.4102570000	1.7266050000	-1.5542240000
C	1.4607320000	-1.5258320000	-0.8965450000
O	2.4102570000	-1.7266050000	-1.5542240000
C	-1.4607320000	-1.5258320000	-0.8965450000
O	-2.4102570000	-1.7266050000	-1.5542240000
C	1.4607320000	1.5258320000	-0.8965450000
O	2.4102570000	1.7266050000	-1.5542240000
C	0.0000000000	-2.7500190000	1.0834870000
O	0.0000000000	-3.7308830000	1.7249410000

C	0.0000000000	2.7500190000	1.0834870000
O	0.0000000000	3.7308830000	1.7249410000

Basis: aug-cc-pVTZ

E_CASSCF(18,13) = -3516.24319296

B. FeCl3

Fe	-0.0000000000	0.0000000000	0.0000000000
Cl	2.1344140475	0.0000000000	0.0000000000
Cl	-1.0672070237	-1.8484567873	0.0000000000
Cl	-1.0672070237	1.8484567873	0.0000000000

Basis: Fe=cc-pVTZ, Cl=cc-pV(T+d)Z

E_CASSCF(9,12) = -2641.23932176

E_CASSCF(15,15) = -2641.25488483

C. Al4O2

Al	1.6519685400	0.0000000000	-0.1783916628
Al	-0.8259842700	-1.4306467250	-0.1783916628
Al	-0.8259842700	1.4306467250	-0.1783916628
Al	0.0000000000	0.0000000000	-2.5146279828
O	0.0000000000	0.0000000000	1.8216083372
O	0.0000000000	0.0000000000	3.3216083372

Basis: aug-cc-pVDZ

E_CASSCF(12,12) = -1117.34334554

Adding Al(3py) as AVAS target orbital of the first Al atom yielded E = -1117.36039759 (see main text).